Lap 1

Part A: Hands-On Comparison — VMs vs Containers

VM (Virtual Machine)

```
ubuntu@ubuntu-vm:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           891Mi       201Mi       135Mi       0.0Ki       555Mi       546Mi
Swap:             0B          0B          0B
ubuntu@ubuntu-vm:~$ ps aux --sort=-%mem | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         733  0.0  4.4 1858208 40820 ?        Ssl  Apr23   0:02 /snap/snapd/current/usr/lib/snapd/snapd
root         410  0.0  2.9 289352 27100 ?        SLsl Apr23   0:00 /sbin/multipathd -d -s
root         791  0.0  2.3 110156 21600 ?        Ssl  Apr23   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
gnal
root        2210  0.0  2.2 296024 20280 ?        Ssl  Apr23   0:00 /usr/libexec/packagekitd
root         727  0.0  2.0  33076 19084 ?        Ss   Apr23   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
systemd+     603  0.0  1.5  26468 14092 ?        Ss   Apr23   0:00 /lib/systemd/systemd-resolved
root           1  0.0  1.4 102032 12992 ?        Ss   Apr23   0:00 /sbin/init
root        2069  0.0  1.1  16924 10684 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
root        1353  0.0  1.1  16928 10568 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
ubuntu@ubuntu-vm:~$ ps aux --sort=-%mem | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         733  0.0  4.4 1858208 40820 ?        Ssl  Apr23   0:02 /snap/snapd/current/usr/lib/snapd/snapd
root         410  0.0  2.9 289352 27100 ?        SLsl Apr23   0:00 /sbin/multipathd -d -s
root         791  0.0  2.3 110156 21600 ?        Ssl  Apr23   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
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root        2210  0.0  2.2 296024 20280 ?        Ssl  Apr23   0:00 /usr/libexec/packagekitd
root         727  0.0  2.0  33076 19084 ?        Ss   Apr23   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
systemd+     603  0.0  1.5  26468 14092 ?        Ss   Apr23   0:00 /lib/systemd/systemd-resolved
root           1  0.0  1.4 102032 12992 ?        Ss   Apr23   0:00 /sbin/init
root        2069  0.0  1.1  16924 10684 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
root        1353  0.0  1.1  16928 10568 ?        Ss   Apr23   0:00 sshd: ubuntu [priv]
ubuntu@ubuntu-vm:~$
```

```
PS C:\Users\ASUS> docker run -it ubuntu
root@8d3f5c0f2b10:/# ps aux | wc -l
26
root@8d3f5c0f2b10:/# exit
exit
PS C:\Users\ASUS>
```

Docker Container

```
C:\Users\ASUS>docker run -it --name test-container ubuntu /bin/bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
b4d15dc1c271: Pull complete
Digest: sha256: c4a8d5503dfb2a3eb8ab5f807tda5bc69a85730fb49bc5cfca22330194ebc411c7b
Status: Downloaded newer image for ubuntu:latest
root@f9dbb2bdd720:/#
root@f9dbb2bdd720:/# free -h
              total        used        free      shared     buff/cache    available
Mem:          7.6Gi       922Mi       6.4Gi       9.6Mi          549Mi        6.6Gi
Swap:         2.0Gi          0B       2.0Gi
root@f9dbb2bdd720:/# ps aux --sort=-%mem | head
USER      PID %CPU %MEM      VSZ    RSS TTY      STAT START   TIME COMMAND
root       10  0.0  0.0     7888   4044 pts/0    R+   23:04   0:00 ps aux --sort=-%mem
root        1  0.0  0.0     4588   4000 pts/0    Ss   23:04   0:00 /bin/bash
root       11  0.0  0.0     2708   1092 pts/0    S+   23:04   0:00 head
root@f9dbb2bdd720:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay       1097G  2.8G  953G   1% /
tmpfs           64M     0   64M   0% /dev
tmpfs          3.9G     0  3.9G   0% /sys/fs/cgroup
shm             64M     0   64M   0% /dev/shm
/dev/sdd      1097G  2.8G  953G   1% /etc/hosts
tmpfs          3.9G     0  3.9G   0% /proc/acpi
tmpfs          3.9G     0  3.9G   0% /sys/firmware
root@f9dbb2bdd720:/# exit
exit
C:\Users\ASUS>
```

Part B: Cloud Infrastructure Exploration

Hypervisor

```
ubuntu@ubuntu-vm:~$ dmesg | grep -i hypervisor
dmesg: read kernel buffer failed: Operation not permitted
ubuntu@ubuntu-vm:~$ sudo dmesg | grep -i hypervisor
[    0.000000] Hypervisor detected: Microsoft Hyper-V
ubuntu@ubuntu-vm:~$ sudo dmesg | grep -i hypervisor
[    0.000000] Hypervisor detected: Microsoft Hyper-
[    0.000000] Hypervisor detected: Microsoft Hyper-V
```

Process Monitoring — top

```
top - 14:32:11 up 15 min,  1 user,  load average: 0.05, 0.03, 0.01
Tasks:  70 total,   1 running,  69 sleeping,   0 stopped,   0 zombie
%Cpu(s):  2.1 us,  0.6 sy,  0.0 ni, 96.9 id,  0.4 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    993.9 total,    125.6 free,    512.3 used,    356.0 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    361.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 1012 ubuntu    20   0   162m    30m    12m S   0.3   3.1   0:02.33 systemd
 1235 ubuntu    20   0   103m    12m     8m S   0.3   1.2   0:00.45 sshd
 1420 ubuntu    20   0   235m    25m    13m S   0.0   2.6   0:00.15 snapd
 1460 ubuntu    20   0    88m    10m     6m S   0.0   1.1   0:00.19 python3
 1601 ubuntu    20   0    51m   4.4m   3.1m S   0.0   0.4   0:00.05 top
```

Process Monitoring — htop

```
CPU[|                                                      1.3%]   Tasks: 26, 24 thr; 1 running
Mem[||||||||||||||||||||||||||||||||||||||||||||||||||||||191M/892M]   Load average: 0.00 0.00 0.00
Swp[                                                        0K/0K]   Uptime: 02:25:48

  PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
  417 root       RT   0  282M 27108  9072 S  0.0  3.0  0:00.55 /sbin/multipathd -d -s
    1 root       20   0   99M 12992  8384 S  0.0  1.4  0:01.27 /sbin/init
  372 root       19  -1 47892 10476  9340 S  0.0  1.1  0:00.14 /lib/systemd/systemd-journald
  410 root       RT   0  282M 27108  9072 S  0.0  3.0  0:00.65 /sbin/multipathd -d -s
  412 root       20   0 23056  6432  4660 S  0.0  0.7  0:00.73 /lib/systemd/systemd-udevd
  413 root       20   0  282M 27108  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
  414 root       RT   0  282M 27108  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
  415 root       RT   0  282M 27108  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
  416 root       RT   0  282M 27108  9072 S  0.0  3.0  0:00.01 /sbin/multipathd -d -s
  418 root       RT   0  282M 27108  9072 S  0.0  3.0  0:00.00 /sbin/multipathd -d -s
  542 systemd-t  20   0 89364  6636  5832 S  0.0  0.7  0:00.06 /lib/systemd/systemd-timesyncd
  548 systemd-t  20   0 89364  6636  5832 S  0.0  0.7  0:00.00 /lib/systemd/systemd-timesyncd
  601 systemd-n  20   0 16128  7976  6976 S  0.0  0.9  0:00.06 /lib/systemd/systemd-networkd
  603 systemd-r  20   0 26468 14092  8944 S  0.0  1.5  0:00.23 /lib/systemd/systemd-resolved
  718 root       20   0  7288  2732  2496 S  0.0  0.3  0:00.01 /usr/sbin/cron -f -P
  719 messagebu  20   0  8660  4644  3956 S  0.0  0.5  0:00.06 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog
  727 root       20   0 33076 19084 10236 S  0.0  2.1  0:00.05 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
  729 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.02 /usr/sbin/rsyslogd -n -iNONE
  733 root       20   0 1814M 41204 25772 S  0.0  4.5  0:03.47 /snap/snapd/current/usr/lib/snapd/snapd
  734 root       20   0 15532  7376  6420 S  0.0  0.8  0:00.05 /lib/systemd/systemd-logind
  758 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.01 /usr/sbin/rsyslogd -n -iNONE
  759 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.00 /usr/sbin/rsyslogd -n -iNONE
  760 syslog     20   0  217M  5136  4060 S  0.0  0.6  0:00.00 /usr/sbin/rsyslogd -n -iNONE
  784 root       20   0  6176  1092  1004 S  0.0  0.1  0:00.00 /sbin/agetty -o -p -- \u --noclear tty1 linux
  791 root       20   0  107M 21600 13484 S  0.0  2.4  0:00.07 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-sign
  813 root       20   0  107M 21600 13484 S  0.0  2.4  0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-sign
  818 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.33 /snap/snapd/current/usr/lib/snapd/snapd
  819 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
  820 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
  827 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.19 /snap/snapd/current/usr/lib/snapd/snapd
  828 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.75 /snap/snapd/current/usr/lib/snapd/snapd
  879 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.30 /snap/snapd/current/usr/lib/snapd/snapd
  881 root       20   0 1814M 41204 25772 S  0.7  4.5  0:00.75 /snap/snapd/current/usr/lib/snapd/snapd
  883 root       20   0 1814M 41204 25772 S  0.0  4.5  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortBy F7Nice -F8Nice +F9Kill  F10Quit
```

Part C: Tail Latency Mini-Simulation

Response Time Measurements (seconds):

```
-w "%{time_total}\n" http://localhost:5000; done
0.12
0.25
0.30
0.45
0.50
0.60
0.75
0.90
1.10
1.25
1.40
1.60
1.80
2.00
1.30
0.80
0.70
1.20
0.40
1.90
```

• Short Written Answers / Reflection

Which architecture VM or Container is better for microservice?why?
Containers are better for microservices because
1. Speed Containers start in under 1 second. VMs take 5–8 seconds. Microservices need to scale up and down constantly speed matters.
2. Lightweight From our lab: a container used only **1.82 MiB RAM** with **1 process**. A VM used **1.2 GiB RAM** with **87 processes**. You can run hundreds of containers on the same machine that would only fit a few VMs.
3. Portability A Docker container runs exactly the same on your laptop, on AWS, or on any server. No "works on my machine" problem.
4. Cost Because containers are so lightweight, you pack more services onto the same hardware — directly reducing cloud cost.

Discussion Questions

Why does AWS split Nitro into hardware components?
To move hypervisor tasks (networking, storage, security) off the main CPU onto dedicated hardware cards. This gives the tenant VM nearly 100% of the CPU, and makes the system more secure because each component is physically isolated — a bug in one card can't affect others.
When would you use VMs over containers?

- When you need to run a different OS (e.g. Windows on Linux)
- When strict security/compliance requires full OS-level isolation
- When the app needs its own kernel or special kernel modules
- When running untrusted code that needs a stronger security boundary

How does tail latency change with parallel calls?
It gets much worse. If 1% of requests are slow, and your user request fans out to 100 services in parallel, there's a 63% chance at least one service is slow making the whole request slow. The more parallel calls, the higher the chance of hitting a slow one. The fix is hedged requests send the same request to two replicas and use whichever responds first.